Vinh Thi Nguyen

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Employment

University of California, Berkeley — Berkeley, CA
Charles B. Morrey, Jr. Math Fellow — Jan 2026 – present

Michigan State University — East Lansing, MI
Postdoctoral Research Associate — Jul 2024 – Dec 2025

Institute of Mathematics, Vietnam Academy of Science and Technology — Hanoi, Vietnam
Researcher — 2013 – 2017

Visiting Positions

Institute for Pure and Applied Mathematics (IPAM) — Los Angeles, CA
Postdoctoral Fellow — Fall 2025
Long program:"*Bridging the Gap: Transitioning from Deterministic to Stochastic Interaction Modeling in Electrochemistry*"

Education

Ph.D. in Mathematics, University of Illinois at Chicago (UIC) — 2024
Advisor: Prof. Roman Shvydkoy

M.A. in Mathematics, Indiana University Bloomington (IUB) — 2019

M.Sc. in Mathematics, Institute of Mathematics, Hanoi, Vietnam — 2012
Advisor: Prof. Nguyen Dong Yen

Bachelor of Education in Mathematics, Thai Nguyen University of Education, Vietnam — 2010

Honors & Awards

- AMS-Simons Travel Grant — 2026-2028
- ICM 2026 Travel Grant for Early Career Mathematicians — 2026
- Damber Singh Tomer and Meena Singh Tomer Endowed International Scholarship, UIC — 2024
- Victor Twersky Memorial Scholarship, UIC — 2022
- University Fellowship, Graduate College, UIC — 2019 –2023
- College of Arts and Sciences Fellowship, IUB — Spr 2018
- Vietnam Education Foundation Fellowship (VEF) (*declined*) — Cohort 2016
- Travel Award: The 3rd HLF, Heidelberg, Germany — Aug 2015
- Honored for Excellent Performance in the Master Program, Hanoi Institute of Mathematics — 2012
- Undergraduate Student Fellowship, Thai Nguyen University of Education — 2007–2010

Research Interests

PDEs, SPDEs, Models of Collective Behavior, Interacting Particle Systems, Duality Theory, Interpolation and Extension Theory, Optimization Theory

Publications & Preprints

11. B. Hambly and V. Nguyen, *Variance-feedback noise in interacting diffusions: shape dynamics and mean-field limits*, preprint, arXiv:2610.20771.
10. V. Nguyen, A. Soffer and T. Vu, *Modified scattering and asymptotics for perturbed one-dimensional cubic NLS*, preprint, arXiv:2609.07044.
9. V. Nguyen and T. Vu, *Stability of finite-batch particle mean-field variational inference beyond strong convexity*, preprint, arXiv:2608.11486.
8. V. Nguyen, *Attainment boundaries and escape rates in asymptotically conic optimization*, preprint, arXiv:2608.10341.
7. V. Nguyen, *Non-attainment of minima in non-polyhedral conic optimization: a robust SOCP example*, preprint, arXiv:2510.00318.
6. V. Nguyen, *Exponential relaxation to equilibrium for a kinetic Fokker-Planck-Alignment equation with force*, preprint, arXiv:2310.16339.
5. V. Nguyen, R. Shvydkoy and C. Tan, *Mean-field limits of deterministic and stochastic flocking models with nonlinear velocity alignment*, Advances in Mathematics, **494**:110929, 2026.
4. V. Nguyen, K. Promislow and B. Wetton, *Quasisteady patterns in interfaces: folding and faceting*, Physica D: Nonlinear Phenomena, **482**:134861, 2025.
3. V. Nguyen and R. Shvydkoy, *Continuous model of opinion dynamics with convictions*, Discrete and Continuous Dynamical Systems **43**(11):4010–4026, 2023.
2. V. Nguyen and R. Shvydkoy, *Propagation of chaos for the Cucker-Smale systems under heavy tail communication*, Communications in Partial Differential Equations **47**(9):1883–1906, 2022.
1. N. T. Vinh, D. S. Kim, N. N. Tam, and N. D. Yen, *Duality gap function in infinite dimensional linear programming*, Journal of Mathematical Analysis and Applications **437**(1):1–15, 2016.

Teaching

University of California, Berkeley

- Math 185: Introduction to Complex Analysis — Fall 2026
- Math 123: Ordinary Differential Equations — Spr 2026

I was a teaching assistant for the following courses:

Michigan State University

- MTH 235. Differential Equations — Spr 2025

University of Illinois at Chicago

- MCS 401. Computer Algorithms I — Spr 2024
- MATH 480. Applied Differential Equations — Fall 2022
- MATH 417. Complex Analysis with Applications — Fall 2022

- MATH 313. Analysis I — Fall 2022
- MATH 165. Calculus for Business — Spr 2021, 2022
- MATH 310. Applied Linear Algebra — Fall 2020, Spr 2024
- MATH 539. Functional Analysis — Fall 2020

Indiana University Bloomington

- MATH-M 365. Intro Probability & Statistics — Fall 2017

University of North Carolina at Chapel Hill

- STOR 445. Stochastic Modeling — Spr 2017
- STOR 113. Decision Models for Business and Economics — Fall 2016

University of Science and Technology of Hanoi

- Analysis and Differential Equations — Falls 2014, 2015

Talks

- Workshop: Synergies between Geometry, Probability, and Computation in High Dimensions, ICERM (*lightning talk*) — Aug 1, 2026
- Southern California Applied Mathematics Symposium
  Loyola Marymount University, Los Angeles, CA — Apr 25, 2026
- Applied PDE Seminar, UC Berkeley — Feb 13, 2026
- Workshop: Recent Advances in Nonlinear Partial Differential Equations
  University of Minnesota, Twin Cities (*lightning talk*) — May 13–17, 2024
- Applied PDE Seminar, UC Berkeley (*online*) — Feb 23, 2024
- Applied Math Seminar, UC Santa Cruz — Jan 08, 2024
- Mini-Symposium: Collective behavior in physical and social network models
  The 2023 SIAM Great Lakes Meeting, Michigan State University, East Lansing, MI — Oct 14, 2023
- International Workshop "Variational Analysis and Application"
  Erice, Sicily, Italy — Aug 28 – Sep 05, 2015
- International Workshop on Nonlinear and Variational Analysis
  Center for Fundamental Science, Kaohsiung Medical University, Taiwan — Aug 2014
- The 8th Congress of the Vietnamese Mathematical Society, Nha Trang, Vietnam — Aug 2013

Conferences & Workshops Attended

- Workshop: Partial Differential Equations for Many Particles Systems, ICERM — Aug 3 - 7, 2026
- ICM 2026, Philadelphia, PA — Jul 23 - 30, 2026
- Riviere-Fabes Symposium on Analysis and PDE, University of Minnesota — Apr 29 – May 1, 2022
- Midwestern Workshop on Asymptotic Analysis, Indiana University Bloomington — Oct 5 – 7, 2018
- The Fourth Chicago Summer School in Analysis, University of Chicago — Jun 19 – 30, 2017

- Workshop: Analysis in Lyon, Université Claude-Bernard Lyon 1, France — Oct 26 – 30, 2015
- The 3rd Heidelberg Laureate Forum, Heidelberg, Germany — Aug 23 – 28, 2015

Services

Referee Work: Journal of Optimization Theory and Applications, Journal of Differential Equations, Communications in Mathematical Sciences, MathSciNet

Organizing:

- Co-organizing (with Keith Promislow) the minisymposium *Mean Field Limits of Particle Dynamics* at the SIAM Conference on Applications of Dynamical Systems (DS25), Denver, Colorado, May 11–15, 2025.
- Co-organizing (with Marjorie K. Drake and Paige Bright) the special session *Nonlinear Constraints: A Catalyst for Creativity in Analysis and its Applications* at the 2025 AWM Research Symposium, University of Wisconsin-Madison, May 16–18, 2025.

Outreach:

- Volunteer for Berkeley mini Math Tournament (Apr 12, 2026)
- Participating in a 1:1 coaching session for an event hosted by the UC Berkeley Emeriti Academy (Feb 2026)
- Judge for the Fellowship for Pre-Graduate Studies in Mathematics, 2025, 2026
- Volunteer to help introduce kids to mathematics outside the standard curriculum at Glencairn Elementary School Math Night (Nov 21, 2024)
- Contributor to the Graduate Opportunities for Women (GROW) 2020 organization